

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

02 Technology Diffusion and Markets

NTU Innovation and
Entrepreneurship

Classification: Restricted



Learning Outcomes

1. Examine the Diffusion of Technology and Market Dominance.
2. Explain the Role of the State in promoting Innovation.
3. Discuss how Marketing Insights can help Innovation.

What is (Technology) Diffusion?

Diffusion is the spread of innovations, through market or non-market channels, from first implementation anywhere in the world to other countries and regions and to other markets and firms.

(OECD 2005)

Question:

Why is Technology Diffusion important to entrepreneurs or corporate entrepreneurs?

Innovation and Consumption

For a successful product, there are the following characteristics:

- the increase in technology capability
- product capability and
- increased level of consumption.

Question:

How many iPads (all versions) have been sold since 2010?



**Newton (sold 80,000)
1993**



**iPad
2010**

Other Factors Affecting the Adoption Rate

- **Perceived Characteristics of Innovation**
 - Relative advantage, compatibility, complexity, trialability, observability, (switching) cost, communicability, social approval, etc.
- **Characteristics of Adopters**
 - People: socioeconomic characteristics, personality variables, communication behaviours
- **Characteristics of Environment**
 - Network connectedness, communication channels, etc.

Diffusion of Innovations

Time

- Time lag between innovation and useable product
 - ❑ for example, mobile phones, driverless cars; drones.
- Industry product development times
 - ❑ for example, aircraft and pharmaceutical products > 10 years

Innovation and the market

- Market vision
- Innovative new products and consumption patterns

Social Process

- The emphasis of this approach is on the roles of opinion leaders and change agents working within social structures and systems on new innovations.



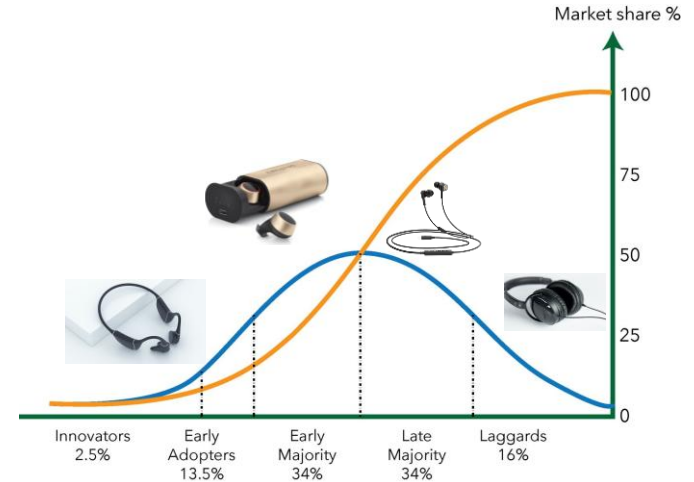
Source: <https://interestingengineering.com/innovation/the-history-behind-the-invention-of-the-first-cell-phone>
accessed on 8 Aug 2023

Innovation Diffusion Theories & Market Share

Explains how an innovation is diffused in a social system over time.

According to Rogers (1962), The category of adopters is based on a normal distribution curve:

- **Innovators** – technology enthusiasts; looking for novelty value
- **Early adopters** – concerned about technology & performance; visionaries; desire to provide opinion leadership
- **Early majority** – pragmatists; deliberate buying decision; willing adopters & followers
- **Late majority** – conservatives; buy because of economic necessity; thinks it's safe to adopt
- **Laggards** – sceptics; traditionalist looking for solutions and conveniences



Source: Rogers, E. (1962) Diffusion of innovations [Adapted from Trott, P. (2021)]

Identify Lead Users



Recognise requirements early



Expect high level benefits from the product



Develop their own innovations and applications



Perceived to be pioneering and innovative.

Question:

Who are the Lead Users of bone-conducting headsets?

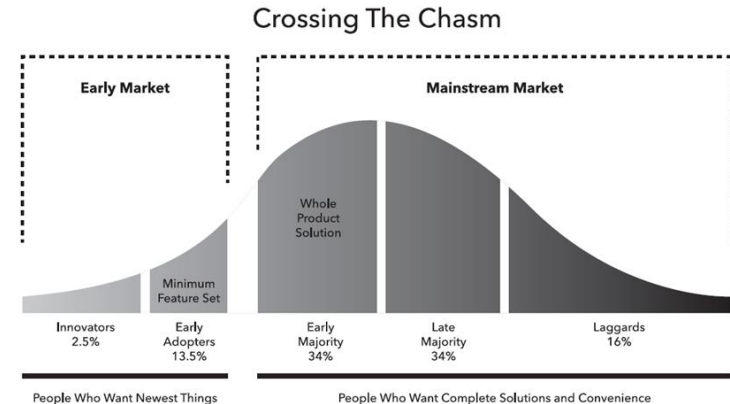
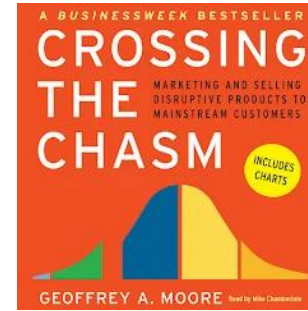
AFTERSHOKZ



Source: <https://us.aftershokz.com/>

“Crossing the Chasm”

- A gap between Early Adopter and Early Majority groups in the Technology Adoption Life Cycle.
- *Credibility gap* that arises from using the group on the left as a reference base for the customer base on the right.
- The chasm exists because consumers trust references from people that belong to their own adopter group.



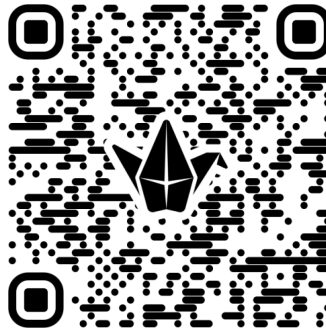
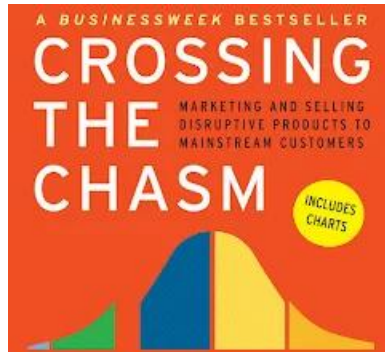
Source: Crossing the Chasm [Adapted from Krawczyk, B. (2022)]

Discussion 2.1

- Which recent successful product/service made use of ***opinion leadership or influencers*** to help the entrepreneurial company gain market traction?



Source: <https://happycall.kr/>
accessed on 9 Jun 2023



Source: <https://thermomix.com.sg> accessed 9 Jun 2023

The Role of the State in Innovation Diffusion

Customer

Lead user

Public procurement

Innovative products and services

Proton beam therapy for cancer to be available in Singapore by early 2023

28 Nov 2022 | The Straits Times

Source: <https://www.nccs.com.sg/news/patient-care/proton-beam-therapy-for-cancer-to-be-available-in-singapore-by-early-2023> accessed on 10 May 2023



Customer

Innovator



Innovator

MOH TraceTogether
Covid-19 App

Source: <https://www.developer.tech.gov.sg/products/categories/digital-solutions-to-address-covid-19/tracetgether/overview.html> accessed on 10 May 2023

Govt

Researcher



Researcher
A*Star Labs

NEWS RELEASE 8-FEB-2023

Breakthrough technology transforms waste plastic bottles into polymers for lithium-ion batteries

Peer-Reviewed Publication

AGENCY FOR SCIENCE, TECHNOLOGY AND RESEARCH (A*STAR), SINGAPORE

Source: <https://www.aurekalert.org/news-releases/979214> accessed on 10 May 2023

Regulator
and policy
maker

Regulator and
Policy Maker
E-scooters

Land Transport Authority
We Keep Your World Moving



Source: <https://www.mobot.sg/blogs/articles/how-to-get-an-ams-or-medical-cert-2026-pma-rules-mobot> accessed on 9 Aug 2026

Question: Does their role promote or inhibit innovation diffusion in the above instances?

Source: Adapted from Trott, P. (2021). *Innovation Management and New Product Development*. 7th edition. Pearson.



The Role of the State – Innovation Policy

Innovation policy has generally focused on four key objectives:

1. The **generation of new knowledge**.
2. **Making government investment in innovation more effective**.
3. Enhancing **diffusion of knowledge and technology** (network interaction effects).
4. Establishing the **right incentives to stimulate private sector innovation** to transform knowledge into commercial success.

The Role of the State – Entrepreneurship Policies

- Concentrate on developing an **environment to foster the emergence of new entrepreneurs and new firms.**
- Yet there has been limited recognition of the full integration of entrepreneurship and innovation.
- Unfortunately, all too often **innovation policies do not incorporate entrepreneurship as a focus.**
- Yet, we know that **entrepreneurs are essential to convert knowledge into economic and social benefits.**

The Role of the State – Governance & Dissemination of Information

Media Releases | Published Date: 17 January 2022

MAS Issues Guidelines to Discourage Cryptocurrency Trading by General Public

Source: <https://www.mas.gov.sg/news/media-releases/2022/mas-issues-guidelines-to-discourage-cryptocurrency-trading-by-general-public#:~:text=The%20Monetary%20Authority%20of%20Singapore,the%20general%20public%20in%20Singapore> accessed on 5 Jan 2023



Monetary Authority of Singapore

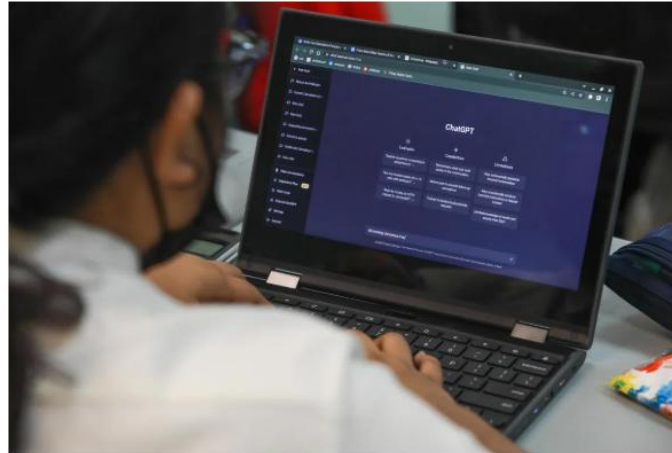
GUIDELINES ON PROVISION OF
DIGITAL PAYMENT TOKEN SERVICES TO THE PUBLIC

Source: <https://www.mas.gov.sg/-/media/MAS-Media-Library/regulation/guidelines/PSO/ps-g02-guidelines-on-provision-of-digital-payment-token-services-to-the-public/Guidelines-on-Provision-of-Digital-Payment-Token-Services-to-the-Public-PS-G02.pdf> accessed on 5 Jan 2023

The Role of the State – Education

All primary, secondary and JC teachers to complete compulsory AI modules under new MOE plan

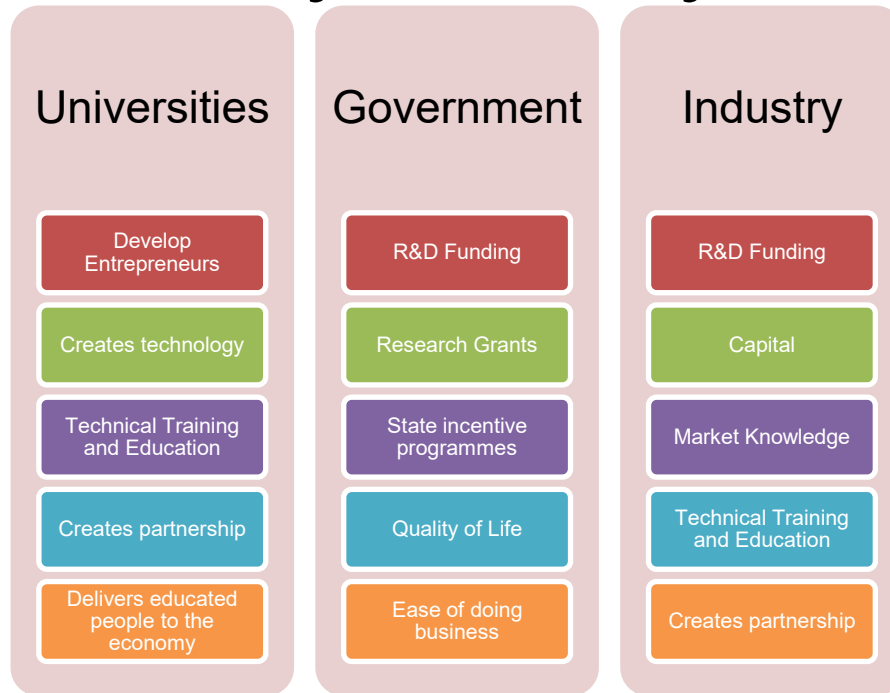
[Sign up now](#): Get tips on how to help your child succeed



A Fuchun Secondary School student using ChatGPT during an English language lesson on April 6, 2023. ST PHOTO: RYAN CHIONG

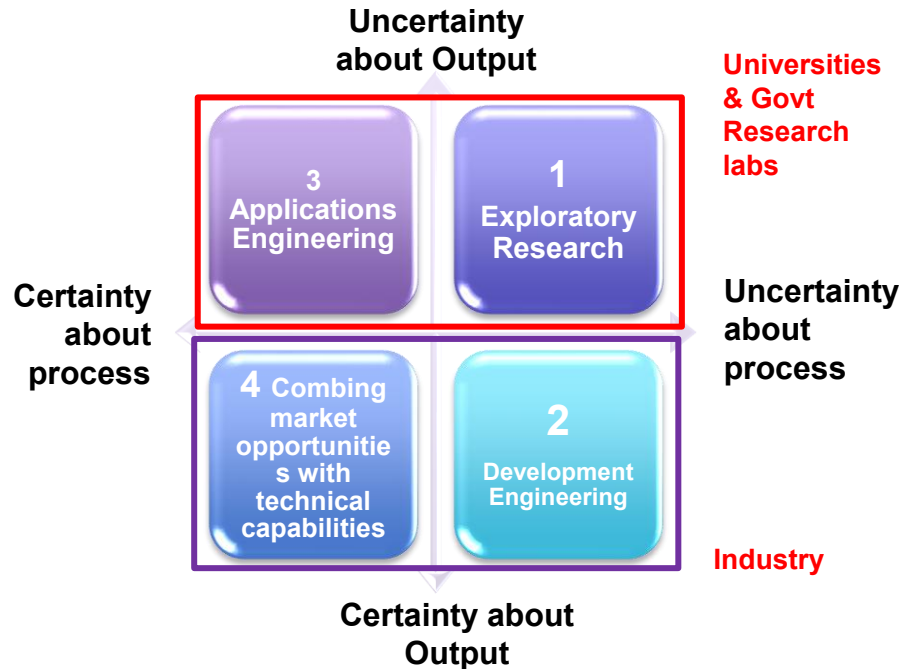
Source: <https://www.straitstimes.com/singapore/parenting-education/all-primary-secondary-and-jc-teachers-to-complete-compulsory-ai-modules-under-new-moe-plan> accessed on 9 Aug 2026

Triple Helix: University-Industry-Government



Source: Adapted from Trott, P. (2021). *Innovation Management and New Product Development*. 7th edition. Pearson.

Managing Uncertainty – Pearson Uncertainty Map



Universities & Govt Research Labs would typically be carrying out Exploration Research, Development Engineering and Applications Engineering.

The Industry would typically be capitalising on combining market opportunities with their technical capabilities, doing some applications and development engineering.

Governments promoting Innovation in their countries are funding both the Universities and the Industry.

Innovation in An Organisation Context

Resources in the Company

Knowledge

Skills

Money

Market
Experience

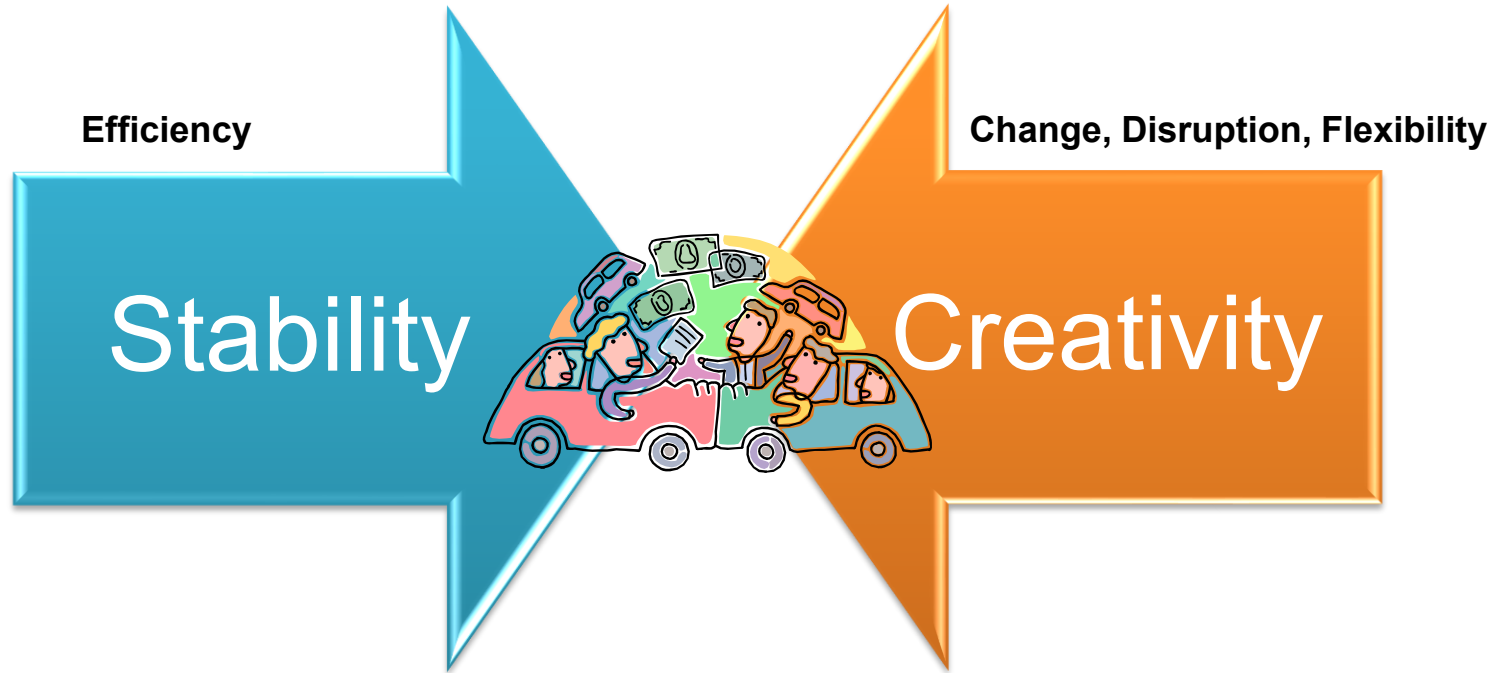
Personalities Required in the Company

Inventor

Entrepreneur

Business
Sponsor

Dilemma of Innovation Management



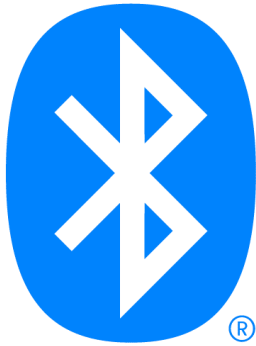
Marketing and R&D

Marketing People about Technical People	Technical People about Marketing People
Have no sense of time!	Want everything right here, right now!
Are interested in technology	Cannot make up their minds
Have a very narrow view of the world	Do not understand technology
Never finish developing a product	Want to ship products before they are ready
...	...

Marketing Insights & Innovation

- What are the potential applications of a technology?
- What application(s) should be pursued first?
- What benefits can the proposed product offer to potential customers?
- What is the potential market size and is this sufficient?
- Can we tap on the wisdom of the customers?
- Can we offer more value with less offerings?

Technology Road Map & New Ideas

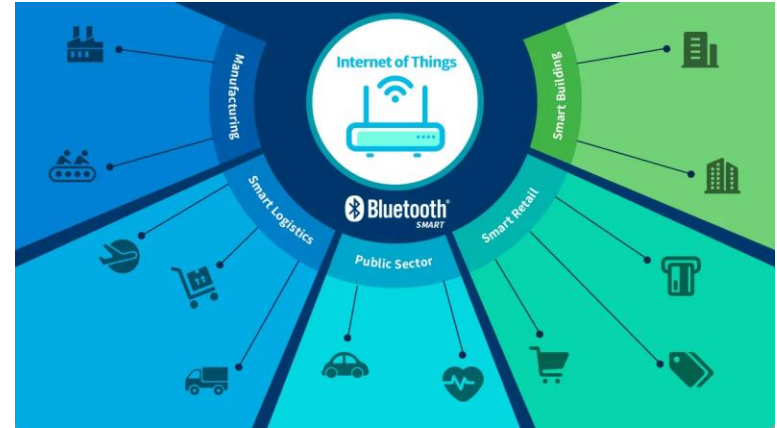


<https://www.bluetooth.com/>

Early days of Bluetooth connectivity use



Future of Bluetooth connectivity use



Source: <https://www.mokoblue.com/why-bluetooth-iot/> accessed on 9 May 2023

Crowdsourcing for new product ideas

1

Ask which product customers would like produced.

2

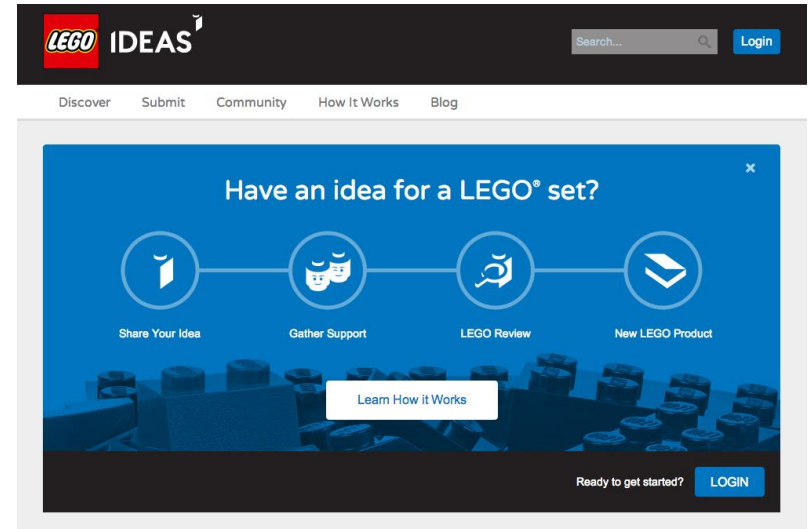
Ask which products customers prefer.

3

Ask customers to name the new product.

4

Ask customers for new product variations.



Source: <http://buytheway.ascjclass.org/how-lego-uses-crowdsourcing-for-new-product-ideas/> accessed on 9 May 2023

Frugal Innovation & New Products

The process of reducing the complexity and cost of a good and its production.

Removing non-essential features from a durable good to sell it in developing countries.

Source: <https://tnfis.wordpress.com/2013/12/08/the-curious-case-of-tata-nano-car/> accessed on 9 May 2023

Building the world's cheapest car

The cheapest car in the world, set to sell for just \$2,500, is being unveiled at the Delhi Auto Expo by the Indian car manufacturer Tata Motors. The "one lakh" – slang for 100,000 rupees – people's car is aimed at the country's 65 million scooter riders currently unable to afford a car

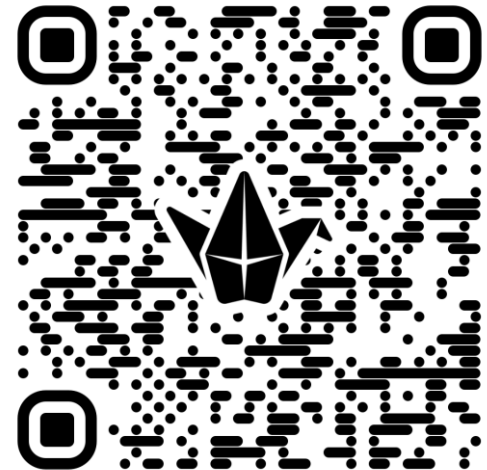


Sources: Industry experts

Discussion 2.2

What products and services had been developed by Entrepreneurial firms as a result of

- Using the Technology Road Map,
- Crowdsourcing of Ideas,
- Frugal Innovation?



Recommended Readings

- Rogers, E.M. (1962). Diffusion of Innovations. Simon & Schuster: 150
- Bluetooth Road Map: <https://www.bluetooth.com/bluetooth-resources/bluetooth-roadmap-building-whats-next/>

End